

Amulya A Shetty

📞 7349219721 ✉ amulyaashetty02@gmail.com [🌐 linkedin.com/amulya-a-shetty](https://www.linkedin.com/amulya-a-shetty) [🐙 github.com/amulyaashetty](https://github.com/amulyaashetty)

Education

PES University

B.Tech in Electronics and Communication

Dec 2020 - May 2024

Bangalore

Experience

Hewlett Packard Enterprise

SDET Intern

Feb 2024 – June 2024

Bangalore

- Worked as an SDET in the Quality Center of Excellence, developing and executing load, performance and automation test scripts for SAP, web, and API applications using NeoLoad, SAP GUI, and Postman.

Flexera Software

Software Engineer

July 2024 – Present

Remote

- Part of the **AI Centre of Excellence (AICOE)**, driving innovation through the development of **Multi Agent Framework** to enable autonomous, context-aware task automation across **FinOps**, **ITAM**, and **CSaaS** domains.
- Worked on **GenAI-powered RAG systems**, **ETL automation using AWS Glue and Databricks**.
- Hands-on with **AWS Bedrock**, **SageMaker**, **LangGraph**, and **Terraform** to design and deploy scalable AI driven solutions across **multi-cloud environments**.
- Developed **interactive Power BI dashboards and visualizations** to track AI model performance, scalability, readiness metrics, and FinOps cost optimization insights.

Projects

Smart Data Hub (SDH) | PySpark, AWS Glue, Step functions, Lambda, Terraform, Docker, Python, PowerBI

- * Contributed to the **Smart Data Hub (SDH)** initiative by developing an AI-driven evidence-to-catalog mapping engine to automate and accelerate the content curation pipeline.
- * Developed scalable **PySpark-based ETL workflows on AWS Glue** to process and map large volumes of software evidence data efficiently.
- * Implemented **CI/CD pipelines using Terraform and Docker**, ensuring reliable, environment-agnostic deployments and automated infrastructure provisioning.
- * Achieved an **85% reduction in manual data mapping effort**, improving processing speed and enabling customer-ready deliverables within 24 hours.

ITAM Assistant Chatbot | AWS Bedrock Agentcore, Python, LLM, SumoLogic

- * Designed conversational AI system using **Agno and Amazon Bedrock AgentCore** to convert natural language into SQL queries for enterprise IT Asset Management with **LLM-as-a-Judge evaluation** and Insight generation.
- * Built a guardrail agent for a defense-in-depth, pre-execution validation system that combined pattern-based SQL injection detection with prompt injection mitigation.

RAG Based License Recommendation for PO | Python, Bedrock Agentcore, Opensearch

- * Designed and built an AI-driven License Recommendation Engine to automatically map purchase orders to software licenses in Flexera One ITAM, improving compliance accuracy and reducing manual effort for customers.
- * Developed **multi-agent orchestration, RAG and MCP** pipelines using **OpenSearch Vector DB** for semantic matching and multi-step reasoning for handling ambiguous or missing purchase data.

Publication and Achievements

- **Hardware Trojan Detection in ISCAS-89 Circuits Using Supervised Machine Learning Algorithms**, A. A. Shetty, in *Proceedings of the 2024 International Conference on Automation and Computation (AUTOCOM)*, 2024.
- **Smart IV Bag Monitoring and Alert System**, A. A. Shetty and Pushpa, in *IEEE Conference Publication*, 2023.
- Won **Flexera Rookie of the year award** for contributions to the engineering team.

Technical Skills

Languages: Python, Scala, SQL

Databases: Oracle, Postgress, MySQL

AI/ML: RAG, MCP, Agentic AI, LangFuse, LangChain

Cloud Platforms: AWS, GCP, Databricks

DevOps: CI/CD, ETL, Terraform, Docker, Vector databases

Core Concepts: Operating System, AI and ML, Database Normalization, Agile Methodology, Cloud Computing, Prompt Engineering

Certifications

AWS Certified Cloud Practitioner (CLF-C02)